

Vaishnav Bengeri

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AI Platform Infrastructure Engineer | DevSecOps Engineer | Active DoD Secret Clearance

Infrastructure, DevSecOps, and AI platform engineer with experience supporting DoD, DHS, and regulated production environments across CI/CD, release engineering, containerization, AWS GovCloud, observability, and software supply-chain security. Trusted with privileged infrastructure access and end-to-end release ownership for mission-focused systems spanning 30+ services/components. Actively building SignalForge AI, an AI-enabled intraday options platform focused on LLM inference, agent workflows, model evaluation, GPU optimization, and cost-efficient AI deployment.

WORK EXPERIENCE

VES — Department of Defense | Artillery Execution Suite (AXS)

Nov 2023 - Present

DevSecOps / Platform Infrastructure Engineer • Full-time

Picatinny Arsenal, NJ (U.S. Army Installation)

- Own secure platform infrastructure and end-to-end release engineering for AXS, a mission-critical DoD software suite with 30+ services/components, spanning source control, CI/CD, security scanning, artifact packaging, containerization, installer generation, and deployments across on-premises and AWS GovCloud environments.
- Serve as one of only three engineers trusted to build and deploy the full AXS product end-to-end, acting as the second-line technical owner for release execution, infrastructure operations, runner availability, secure delivery workflows, and production-ready deployment readiness.
- Design, optimize, and maintain GitLab CI/CD pipelines across 30+ active projects, automating testing, integration, security scanning, artifact generation, packaging, and deployment workflows; reduced deployment time by 30% within 3 months.
- Administer multiple mirrored GitLab environments supporting controlled developer contribution workflows, authoritative source control, secure code custody, release coordination, and infrastructure-aligned software delivery within a DoD-controlled environment.
- Operate platform observability and application monitoring workflows using Grafana for infrastructure/resource monitoring and ELK Stack for application performance monitoring, log analysis, troubleshooting, and deployment validation.
- Manage privileged operations for on-premises Linux runners, AWS GovCloud EKS workloads, and EC2-hosted GitLab runners, including runner provisioning, access-controlled administration, resource monitoring, troubleshooting, capacity management, and release-blocker resolution.
- Engineer reproducible Gradle build automation for all AXS components, supporting dependency consistency, controlled artifact generation, reliable packaging, and repeatable release execution across the full platform.
- Build and maintain Dockerized versions of the AXS platform across 30+ services/components, improving portability, environment consistency, deployment reliability, and operational readiness across on-premises, Kubernetes, and GovCloud infrastructure targets.
- Implement multi-layer application and software supply chain security controls using Trivy, ClamAV, TruffleHog, Semgrep, SonarQube, CI/CD scanning gates, and privileged runner hardening to detect vulnerabilities, malware, secrets, code quality issues, and dependency risks before release.
- Supported DevSecOps, build, deployment, and infrastructure workflows for additional DoD IRAD efforts including X-VIRT and DobOS, extending secure delivery practices across adjacent research and development initiatives.

IBM — Department of Homeland Security | FEMA Risk MAP

Jun 2022 - Oct 2023

Application Developer II / I • Full-time

Rocket Center, WV

- Performed DevSecOps, platform infrastructure, and release engineering responsibilities for FEMA Risk MAP under the Department of Homeland Security, supporting secure application delivery, CI/CD automation, Kubernetes deployments, cloud migration, and production release operations across government-managed environments.
- Designed, maintained, and deployed CI/CD pipelines using GitLab, Jenkins, Ansible Tower, Groovy shared libraries, and Agile orchestration tools including JIRA and IBM Rational Team Concert, streamlining build, test, deployment, and release workflows for multiple applications.
- Developed Kubernetes Helm charts, YAML configurations, Dockerfiles, and containerized application images to deploy and manage application/service workloads across self-managed and AWS EKS-managed Kubernetes clusters, including public and internal traffic routing.
- Helped lead migration efforts from legacy development and deployment workflows to modern cloud-native platform infrastructure using AWS EC2, AWS EKS, S3, GitLab, Nexus, Git LFS, Terraform, Helm, and containerized delivery patterns.

- Managed production-privileged credentials, government-managed servers, and deployment access while facilitating production releases for containerized and legacy applications, including release readiness, environment validation, operational troubleshooting, and secure delivery execution.
- Developed Python, PowerShell, and Groovy automation scripts to support batch operations, administrative tasks, Jenkins shared libraries, deployment repeatability, and CI/CD standardization across multiple projects.

EDUCATION

Bachelor of Science in Computer Science, Minor in Mathematics

Rutgers University – New Brunswick

New Brunswick, NJ • Sep 2018 - May 2022

CERTIFICATIONS

CompTIA Security+

Nov 2025 - Nov 2028

AWS Certified Machine Learning Engineer - Associate (In Progress)

Oct 2026

PROJECTS

SignalForge AI

Mar 2026 - Present

AI-Enabled Intraday Options Research Platform

- Building an intraday options trading research platform with scanner automation, candidate history, simulated paper-trade approvals, outcome tracking, risk dashboards, and AI-assisted analysis. The repository describes the platform flow as scan → candidate history → trade approval → simulated paper trades → outcomes → risk dashboard → AI analysis.
- Implemented an AI inference layer that generates approval-time trade analysis with structured outputs including bull/bear cases, liquidity concerns, quote/spread quality concerns, suggested action, confidence, and post-approval monitoring notes.
- Expanding the platform’s AI footprint through Postgres-backed inference/tool-call traces, agentic scanner-to-decision workflows, historical replay mode, eval harnesses, prompt/agent versioning, CI-gated eval promotion, dashboard trace viewing, alternative-model comparisons, and self-hosted model cost-vs-quality evaluation.
- Engineering GPU optimization and model cost-efficiency improvements to support scalable AI inference, experiment evaluation, and future self-hosted model deployment.

Elite Options Scanner

Dec 2025 - Present

Gateway Scanner for SignalForge Platform

- Developed a Python-based single-leg options scanner that uses Tradier when available and falls back to yfinance, evaluating contracts by liquidity, directional regime alignment, and estimated expected value.
- Built configurable scanner logic for spread, open interest, volume, moneyness, budget, expiration, and risk filters, with CSV export support for research workflows.
- Evolved the scanner into the gateway component for SignalForge AI, feeding ranked options candidates into the broader research, approval, simulated paper-trading, monitoring, and AI analysis platform.

SKILLS

AI / ML: agentic workflows, AI inference pipelines, context engineering, cost-vs-quality analysis, eval harnesses, GPU inference optimization, LLM application development, OpenAI Platform API, tool-calling

Cloud & Infrastructure: AWS EC2, AWS EKS, AWS GovCloud, AWS S3, Google Cloud, Linux, on-premises infrastructure, privileged infrastructure operations, Red Hat Enterprise Linux, Windows Server

Containers & Orchestration: containerized deployments, Docker, Docker Compose, EKS workloads, Helm, Kubernetes, OCI-compliant images, service routing, YAML

CI/CD & Automation: Ansible Automation Platform, Ansible Tower, GitHub, GitHub Actions, GitLab CI/CD, GitLab runners, Gradle, Groovy, Jenkins, Palantir Apollo / Rubix, PowerShell, Python, Terraform

Security & DevSecOps: ClamAV, dependency risk detection, malware scanning, OWASP, secrets detection, Semgrep, software supply-chain security, SonarQube, static analysis, Trivy, TruffleHog, vulnerability scanning, Zap

Monitoring & Observability: application performance monitoring (APM), Elasticsearch, ELK Stack, Grafana, infrastructure monitoring, Kibana, log analysis, Logstash